PROGRAMMING FOR BUSINESS REPORT

ANALYSING COVID-19 VACCINATION TRENDS

BUSN5101 Programming for Business

Group 33

Indumathi Ramalingam - 24558135
Shruthikaa Sundarrajan - 24455168
Isha Vanarase - 24613719
Tagur Kodati - 24541764

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Introduction

The Coronavirus Disease -19 pandemic shaped the whole world impacting both health systems and economies. The development of vaccines was the most important measure in tackling the pandemic. Due to vaccines the spread of the virus was reduced, enabled businesses to reopen, and helped economies recover. Despite the success of the vaccines the distribution of vaccines has its obstacles related to logistics and unequal access in various regions. With the help of a dataset from Our World in Data, the report analyses the trends of global vaccination rates and provides valuable insight related to the success and differences in vaccine distribution.

The report aims to examine the relationship between vaccination rates in different countries and find out the key patterns that can guide businesses to adapt according to uncertain public health situations. Therefore this report will help understand the trends to make a data-driven approach for organisations to have sustainable growth in the post-pandemic world.

Background

The COVID-19 pandemic impacts the global health and economic environments which highlights the importance of developing effective vaccines. These vaccines are vital tools to mitigate the virus's spread and reduce severe cases.

The distribution of vaccines is directly linked with economic recovery and growth. According to a study in the Swiss Journal of Economics and Statistics, the rollout of COVID-19 vaccines has boosted consumer confidence, enabled businesses to reopen, and highlighted the importance of sharing doses to increase global economic activity like trade (Deb et al., 2022).

Even though these vaccines showed successful results it has its challenges. The COVID-19 vaccine distribution challenges were due to supply chain issues, public hesitancy, and uneven access (Ariyarajah et al., 2022). Therefore, these differences affected the efforts for vaccination and also the economic recovery.

The dataset used for the report is from “Our World in Data” which tracks the number of fully vaccinated individuals per hundred throughout different countries. This dataset helps in comparing the nations with daily vaccine and booster rates to understand the relationship between vaccination rates and identify trends of vaccination roles in different countries.

Methodology

This report aims to analyse and visualise the number of COVID-19 vaccinations produced and used around the world.

To start with, pandas and matplotlib libraries were imported for data analysis and visualisation and the vaccination data was read. After the vaccination data was loaded into a data frame and initial rows in the dataset were viewed as an overview. Before starting with data analysis, the data was cleaned to remove any irrelevant data like world and continent names. In the cleaning process, the first few rows of the dataset were viewed to understand the structure. The unique locations were identified in the “location” column of the data to determine and remove any irrelevant entries. Once the unnecessary locations were removed, the “location” column was renamed to “countries” for better understanding.

After cleaning the dataset, we analysed the top and bottom five countries for fully vaccinated individuals in 2021 and 2022 to understand the trends over these two years. We also identified the date with the highest number of vaccinations given and narrowed down the analysis to the top and bottom five countries with the highest number of vaccinations given. To visualise the vaccination trends from 2020 to 2022, we plotted two line graphs, one displaying the trends for the top eight countries and another for the bottom ten countries. Throughout this process, many key concepts from our lectures and seminars were applied.

To analyse and visualise the data different concepts were used that were taught in the lectures and seminars. Following are some of the major concepts that were used in this analysis:

- Out of these concepts, the Pandas library was important to handle this big dataset in tabular format. We imported and used pandas for our data analysis.(Reed, 2024, Slide 4)
- To simplify the analysis, the groupby() function was used to group the data by date and get the total number of vaccinations per day. (Reed, 2024, Slide 5).
- To rank countries by vaccination rate, sort_values() function was used for sorting the data in ascending and descending order to get the top and bottom five countries in vaccinations. (Reed, 2024, Slide 4)

- The concept of function block is used twice in this analysis. It calculates the top and bottom five countries by vaccination rates for specific years and avoids repeated tasks like sorting. The function block was also used to create line charts for these countries without repeating the chart code. (Reed, 2024, Slide 3)
- Data visualisation played an important role in analysing and summarising the results. The import of Matplotlib library was crucial and relevant for visualisation in this report.(Reed, 2024, Slide 5)

Results and Discussion

Top and Bottom Countries for Fully Vaccinated People (Per 100 People) 2021 and 2022

The vaccine dataset for December 31st of each year was filtered to identify the top and lowest 5 countries for 2021 and 2022. In 2021, the top 5 countries were Brunei, Cuba, Chile, Portugal, and Singapore while the bottom 5 countries were Côte d' Ivoire, Guinea, Zambia, Malawi, and Nigeria which shows 89% of Bruneians were fully immunised, but only 2% of Nigerians were. In 2022, the top 5 countries were Singapore, Hong Kong, Chile, Malta, and South Korea while the bottom 5 countries were Timor, Poland, Croatia, Russia, and Kyrgyzstan which results in Singapore and Hong Kong being in the top of the list with 91%, while Kyrgyzstan was at the bottom with 21%.

```
Top 5 Countries in 2021:
   country  people_fully_vaccinated_per_hundred
   Brunei                                  89.0
     Cuba                                  86.0
    Chile                                  84.0
 Portugal                                  83.0
Singapore                                  83.0
```

```
Bottom 5 Countries in 2021:
       country  people_fully_vaccinated_per_hundred
Cote d'Ivoire                                    8.0
       Guinea                                    7.0
       Zambia                                    6.0
       Malawi                                    3.0
      Nigeria                                    2.0
```

```
Top 5 Countries in 2022:
     country  people_fully_vaccinated_per_hundred
  Singapore                                  91.0
  Hong Kong                                  91.0
      Chile                                  90.0
      Malta                                  88.0
South Korea                                  86.0
```

```
Bottom 5 Countries in 2022:
    country  people_fully_vaccinated_per_hundred
     Timor                                  59.0
    Poland                                  57.0
   Croatia                                  56.0
    Russia                                  55.0
Kyrgyzstan                                  21.0
```

It shows that countries like Brunei, Singapore, Hong Kong, and Cuba have high vaccination rates because of strong governance and better healthcare systems. These countries give importance to public health spending and resources are available for vaccination campaigns. On the other hand, due to limited healthcare budgets and logistical constraints countries such as Guinea, Zambia, Kyrgyzstan, and Malawi faced challenges. This finding is based on (Chauhan et al., 2023) which states that nations with higher healthcare expenditure had experienced smoother vaccine rollouts.

Date of Highest Number of Vaccinations Given

```
The date with the highest number of vaccinations given worldwide is: 2021-06-27 and number of vacinations given is: 44043325.0
```

For the 'date' column a readable format is created before data sorting. To determine the total number of vaccinations administered daily the data is categorised by date. By using, group by, sum, and max functions, the day with the highest number of vaccines is identified. The most vaccinations 44043325 doses were given on June 27, 2021, according to the data. This is a major turning point in the global immunisation campaign, this peak shows the increase in vaccine production and the tremendous effort made at the time to combat the disease.

Countries with the Most and Least Vaccinations Given on June 27, 2021

The countries with the highest immunisation rates as of June 27, 2021, were China (1.19 billion) United States (337 million), India (318 million), Brazil (96.79 million), and the United Kingdom (77.04 million) while the lowest was Saint Lucia (52,355), Armenia (74,714), Jersey (122,930), Aruba (125,290), and Namibia (141,209).

```
Top 5 Countries on 2021-06-27 with the most vaccinations:
       country  total_vaccinations
         China        1185787000.0
 United States         337692191.0
         India         318030412.0
        Brazil          96789753.0
United Kingdom          77038257.0
```

```
Bottom 5 Countries on 2021-06-27 with the least vaccinations:
    country  total_vaccinations
Saint Lucia            52355.0
    Armenia            74714.0
     Jersey           122930.0
      Aruba           125290.0
    Namibia           141209.0
```

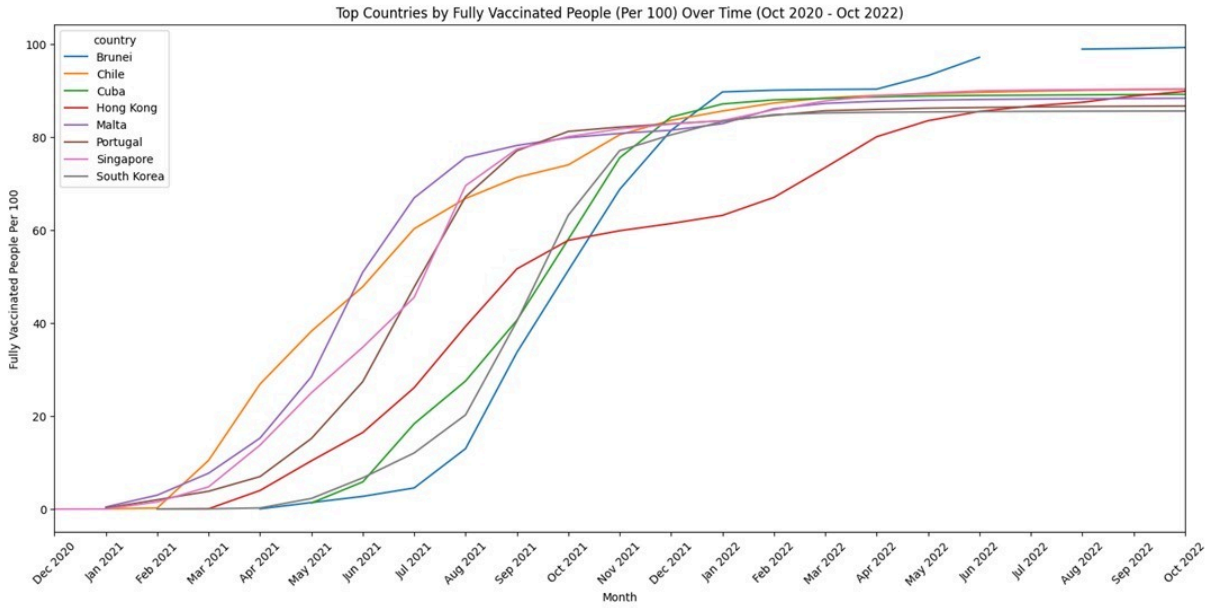
Immunisation rates rose globally compared to 2022 but the distinction between nations with high and low vaccination rates is in higher contrast. The study of the impact of geopolitical socio-economic factors on vaccine dissemination trends: a case study on COVID-19 vaccination strategies points out that due to low healthcare spending and dependence on external aid affect the vaccination rates in the poorest countries. This shows that countries such as India, China, and the U.S. are top in vaccination numbers due to large populations and the capacity of large-scale healthcare operations whereas smaller countries like Saint Lucia and Armenia rely on external vaccine donations.

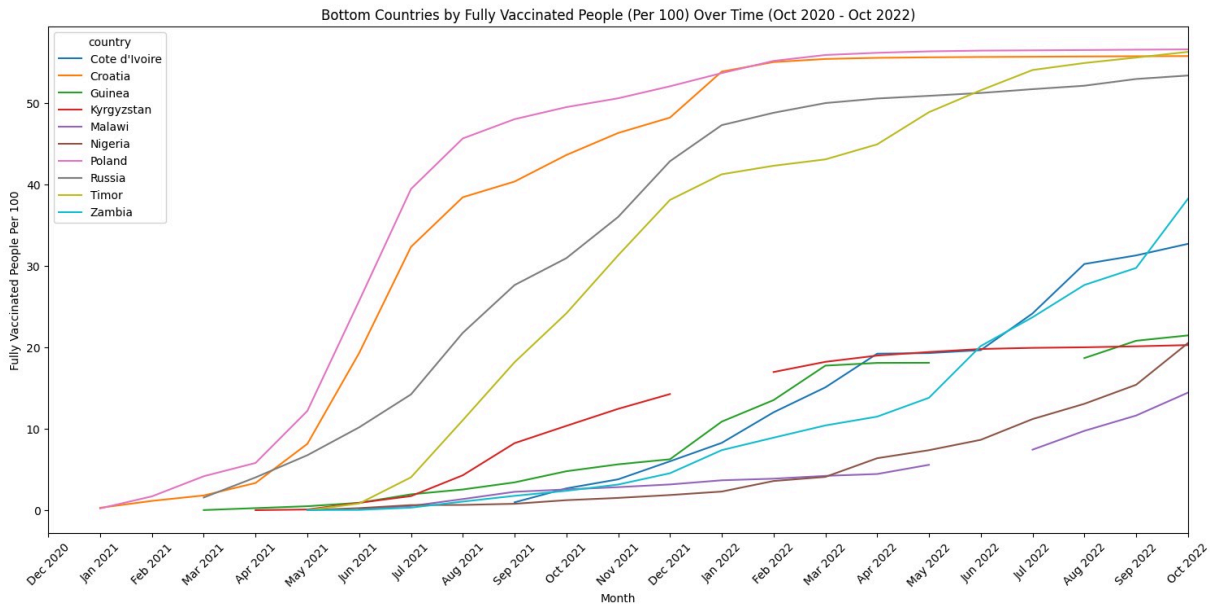
Additionally, countries with low vaccination rates face obstacles due to logistical arrangements and limited resources which aligns with the findings from the study Identifying Priority Challenges and Solutions for COVID-19 Vaccine Delivery in Low- and Middle-Income Countries: A Modified Delphi Study (Ariyarajah et al., 2022), which highlights the restrictions the countries face in vaccine distribution.

Trends over time: Visualisation of the top 10 and bottom 10 countries over the year 2020 - 2022

The visualisation is displayed with the help of a line graph because it effectively conveys the time-based nature of the dataset, showing vaccination trends throughout the given period. The matplotlib is used for two charts to show the changes in vaccination rates over the period from October 2020 to October 2022 for most and least developed nations. The visualisation of each figure therefore shows the trend of vaccination in each country over two years separated by lines for the different countries. This makes a comparison of the rates at which these countries have accepted immunisation over time easier.

In the top countries chart, the countries show a rapid increase in vaccination rates and reach high levels (80 - 100%) and this is followed by a plateau which shows a successful distribution but a hold-up in vaccinating the remaining population.





The bottom countries chart shows the country's vaccination rate is slower but gradually increased at the level of 40 - 50% and it also displays flatlines reflecting the issues discussed above such as supply chain obstacles, social resistance, and more. These graphs very clearly show which countries quickly reached high vaccination levels and which lagged.

Limitations

For effective results, steps were taken to clean and format the dataset for the analysis process and improve data clarity. The dataset contained unnecessary information, including categories such as continents (e.g., Asia, Europe, World) and income groups (e.g., Low income, High income). Therefore these entries were removed from the dataset to focus on the countries to have correct insights.

Additionally, there are many missing values across the columns, which was difficult to further analyse. These gaps can lead to incomplete interpretations and affect the overall quality of the results. The presence of missing data in fields may prevent the application of certain analytical techniques and reduce the effectiveness of data-driven decisions.

Specifically, these null values are consistent for certain months. When attempting to generate a line plot, the missing data results in breaks within the line graph, making it challenging to

visualise the trends accurately. Such inconsistencies can mislead stakeholders and hide meaningful patterns that may exist in the data.

Finally, to improve clarity and understanding, the column name 'location' has been changed to 'country.' This modification enhances the dataset's readability and aligns the terminology with common practices in data analysis. By using 'country,' users can more easily interpret the data and make informed conclusions.

Recommendations and solutions

COVID-19 has caused a major influence on the global business environment, based on these vaccination trends businesses can make strategic decisions. The results showed noticeable differences in the high vaccination and low vaccination rates between the countries. For example, in 2021, Brunei and Cuba had 89.0% and 86.0% of the population fully vaccinated and with the highest vaccination rates while Nigeria had only 2.0% of people vaccinated. This contrast shows the efficiency of healthcare systems in top countries and challenges like logistical constraints and vaccine hesitancy faced by lower countries (Ariyarajah.,2022).

With these differences in vaccination rates, organisations can analyse the trends when deciding on market entry or expansion. For instance, Singapore and Chile have higher vaccination rates showing consumer confidence and economic stability giving opportunities for businesses to expand. On the other hand, countries like Nigeria and Malawi have lower vaccination rates where businesses can adopt strategies to build trust and address their health concerns.

The companies can also change their marketing strategies according to these local vaccination trends by focusing on health and safety messaging, showing their commitment to public health, and providing flexible supply chain management of vaccines. Therefore, investing in logistics technologies that allow real-time inventory tracking is effective for businesses to respond according to the market conditions.

Moreover, companies can partner with local health organisations to improve vaccine access in their Corporate Social Responsibility (CSR) activities related to community help. They can perform by supporting vaccination drives, funding public health campaigns, or providing

resources for healthcare facilities. These initiatives will enhance the organisation's brand reputation and demonstrate a commitment to public welfare which increases the customer's loyalty.

This data-driven decision-making approach will allow the companies to monitor the vaccination trends to adjust their strategies to changes in the market. Therefore, companies can prepare crisis management plans which include vaccination progress to ensure that operations are continuing even during public health emergencies.

Ultimately following these recommendations, businesses can improve resilience, consumer safety, and market positioning in emergencies. These strategies not only help companies to enhance performance but also contribute towards public health through CSR activities which will increase the goodwill of the company. Hence, understanding these vaccination trends plays an important role in shaping successful business strategies in the post-pandemic world.

Conclusion

This report is a review of the vaccination data, covering the COVID-19 global vaccination results based on vaccination rate analysis and public health outcomes. The dataset was organised and looked for vaccination trends over two years, showing key dates for when the most vaccinations were given. The analysis shows large differences in vaccination rates where the areas with better infrastructure attained a comparatively high level of immunisation, while the other areas faced several difficulties that included limited resources and an inability to have proper logistical arrangements.

Even though there are such limitations as missing data and irrelevant categories, the data gives some idea of the value. Overall, results show business strategy aligned to the local trends of vaccination that is entry to particular markets, enhancement of logistics, rebuilding of consumer confidence, and increasing goodwill.

This data-driven approach guides the ability of business operations to adjust to the crisis and sustain continuity in the light of community well-being. This dataset permits an understanding of the vaccination trend to make informed decisions and help build a healthy future.

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Appendices

Appendix A: Pseudocode

```
(IMPORT pandas)
(IMPORT matplotlib.pyplot)
FUNCTION : function(year)
ACTION:
    dataset_years <- FILTER task1_dates WHERE df_filter [ date ] IS year
    data_sort <- SORT dataset_years BY people_fully_vaccinated_per_hundred DESCENDING
    top_5 <- GET FIRST 5 ROWS OF data_sort [[ country , people_fully_vaccinated_per_hundred ]]. ROUND OFF TO 0
    bottom_5 <- GET LAST 5 ROWS OF data_sort [[ country, people_fully_vaccinated_per_hundred ]]. ROUND OFF TO 0

FUNCTION : plot (countries, title)
ACTION:
    country_data <- FILTER monthly_vaccination WHERE 'country' IS IN countries
    pivot_data <- PIVOT country_data WITH 'month' AS INDEX, 'country' AS COLUMNS, 'people_fully_vaccinated_per_hundred' AS VALUES
    ax <- CREATE LINE PLOT OF pivot_data WITH figsize=(12,8)
    TITLE OF PLOT TO title
    X-LABEL TO Month , Y-LABEL TO Fully Vaccinated People Per 100
    ROTATE X-TICKS 45 DEGREES
    REMOVE GRID FROM PLOT
    EXECUTE plt.tight_layout()
    EXECUTE plt.show()
START
dataset <- READ CSV
PRINT FIRST 5 ROWS of dataset
```

```
PRINT SUM OF NULL VALUES IN dataset
 unique_locations <- GET UNIQUE VALUES FROM dataset
PRINT unique_locations
locations <- ['Africa', 'Asia', 'Europe', 'European Union', 'High income', 'Low income', 'Lower middle income', 'North America', 'Oceania', 'South America', 'Upper middle income', 'World'
dataset_cleaned <- REMOVE ROWS WHERE dataset [location] IS IN locations
PRINT dataset_cleaned
RENAME COLUMN dataset_cleaned[ location ] TO country
PRINT FIRST 5 ROWS dataset_cleaned

// Task 1
dataset_cleaned[ date ] <- CONVERT dataset_cleaned[date] TO DATETIME
task1_dates <- FILTER dataset_cleaned WHERE dataset_cleaned[ date ]  IS IN ['2021-12-31', '2022-12-31']
task1_dates<- DROP ROWS WHERE people_fully_vaccinated_per_hundred IS NULL

CALL function ( 2021-12-31 ) <- top_2021, bottom_2021
CALL function ( 2022-12-31 ) <- top_2022, bottom_2022
PRINT  ("Top 5 Countries in 2021: " top_2021)
PRINT ("Bottom 5 Countries in 2021: "  bottom_2021)
PRINT  ("Top 5 Countries in 2022: "  top_2022)
PRINT  ("Bottom 5 Countries in 2022: "  bottom_2022)

 //Task 2
dataset_cleaned['date'] <- CONVERT dataset_cleaned [ date ] TO DATETIME
dataset_cleaned <- SORT dataset_cleaned BY  date
daily_totals <- GROUP BY dataset_cleaned [date] AND SUM daily_vaccinations
max_date <- GET INDEX OF MAX VALUE IN daily_totals
max_vaccinations <- GET MAX VALUE OF daily_totals
 PRINT  ("The date with the highest number of vaccinations given worldwide is:" max_date "and number of vaccinations given is:" max_vaccinations )
```

```
// Task 3
dataset_max_vaccinations <- FILTER dataset_cleaned WHERE dataset_cleaned['date'] IS '2021-06-27'
top_5 <- SORT dataset_max_vaccinations BY total_vaccinations  DESCENDING AND FIRST 5 ROWS
bottom_5 <- SORT dataset_max_vaccinations BY 'total_vaccinations' ASCENDING AND FIRST 5 ROWS
PRINT  ("Top 5 Countries on 2021-06-27 with the most vaccinations:" top_5)
PRINT  ("Bottom 5 Countries on 2021-06-27 with the least vaccinations:" bottom_5)

// Task 4: Create visualisations for top and bottom countries from Task 1
top_countries <- ['Brunei', 'Cuba', 'Chile', 'Portugal', 'Singapore', 'Hong Kong', 'Malta', 'South Korea']
bottom_countries <- ['Nigeria', 'Malawi', 'Zambia', 'Guinea', "Cote d'Ivoire", 'Kyrgyzstan', 'Russia', 'Croatia', 'Poland', 'Timor']
dataset_cleaned['date'] <- CONVERT dataset_cleaned [date] TO DATETIME
dataset_cleaned['month'] <- CONVERT dataset_cleaned [date] TO MONTH-YEAR FORMAT
monthly_vaccination <- GROUP BY [month, country] AND CALCULATE MEAN OF people_fully_vaccinated_per_hundred
monthly_vaccination <- FILTER WHERE month >= 2020-10 AND month <= 2022-10
CALL plot (top_countries, Top Countries by Fully Vaccinated People (Per 100) Over Time (Oct 2020 - Oct 2022))
CALL plot (bottom_countries, Bottom Countries by Fully Vaccinated People (Per 100) Over Time (Oct 2020 - Oct 2022))
END
```

Appendix B: Python Code

```
#importing all necessary libraries
import pandas as pd #for for data handling
import matplotlib.pyplot as plt  #for visualisation

dataset = pd.read_csv(vaccinations.csv") #reading the csv file
print(dataset.head()) #print first five rows
print(dataset.isnull().sum()) #checking for null values

#data cleaning
unique_locations = dataset['location'].unique()  #for checking all unique values of locations
print(unique_locations) #for viewing
locations = [
    'Africa', 'Asia', 'Europe', 'European Union',
    'High income', 'Low income', 'Lower middle income',
    'North America', 'Oceania', 'South America',
    'Upper middle income', 'World']  #found these unnecessary locations for our analysis
dataset_cleaned = dataset[~dataset['location'].isin(locations)]  #removing those unnecessary
locations from above list
print(dataset_cleaned) #now this contains only the country names which will be use full for
analysis
dataset_cleaned.rename(columns={'location': 'country'}, inplace=True) # renaming the column
name; location to country for better understanding
print(dataset_cleaned.head()) # for checking cleaned data ie first 5 rows

#task 1
dataset_cleaned['date'] = pd.to_datetime(dataset_cleaned['date']) #converting the date to datetime
format, where py code can read these date objects
task1_dates = dataset_cleaned[dataset_cleaned['date'].isin(['2021-12-31', '2022-12-31'])] #
checking the required dates whether they are in the columns
task1_dates = task1_dates.dropna(subset=['people_fully_vaccinated_per_hundred']) # it will
check for null values and drops them
```

```
def function(year):      #function block
    dataset_years = task1_dates[task1_dates['date'] == year]  #checking the specified date
    dataset_sort = dataset_years.sort_values(by='people_fully_vaccinated_per_hundred',
ascending=False) #this will sort values from largest to smallest

    top_5 = dataset_sort[['country', 'people_fully_vaccinated_per_hundred']].head(5).round(0)
#for getting the top 5 countries
    bottom_5 = dataset_sort[['country', 'people_fully_vaccinated_per_hundred']].tail(5).round(0)
#for getting the bottom 5 countries

    return top_5, bottom_5 #returning the function

top_2021, bottom_2021 = function('2021-12-31') #applies the function block to the date
mentioned
top_2022, bottom_2022 = function('2022-12-31')
#outputs the values and index=False indicates that it will not show index values of the output
since we dont require that
print("Top 5 Countries in 2021: " + str(top_2021))
print("Bottom 5 Countries in 2021: "+ str(bottom_2021))
print("Top 5 Countries in 2022: " + str(top_2022))
print("Bottom 5 Countries in 2022: " + str(bottom_2022))


#task 2
dataset_cleaned['date'] = pd.to_datetime(dataset_cleaned['date'])
dataset_cleaned = dataset_cleaned.sort_values(by='date')
daily_totals = dataset_cleaned.groupby('date')['daily_vaccinations'].sum() #adds the values of
same date
max_date = daily_totals.idxmax().date()  #for getting the date where vaccinations is max
max_vaccinations = daily_totals.max()   #for getting the total count
#outputs the max vaccinations with the date
print("The date with the highest number of vaccinations given worldwide is:" + str (max_date)+
"and number of vacinations given is:" +str (max_vaccinations))
```

```
#task 3
pd.set_option('display.float_format', '{:.0f}'.format)
dataset_max_vaccinations = dataset_cleaned[dataset_cleaned['date'] == '2021-06-27'] #now
checking the data on 27 june 2021
#sorting values by total vaccinations from max to min value which helps to get top values
top_5 = dataset_max_vaccinations.sort_values(by='total_vaccinations',
ascending=False).head(5)[['country', 'total_vaccinations']].round(0)
#sorting values by total vaccinations from min to max value which helps to get bottom values
bottom_5 = dataset_max_vaccinations.sort_values(by='total_vaccinations',
ascending=True).head(5)[['country', 'total_vaccinations']].round(0)
#outputing the top 5 and bottom 5 countries on 27 june 2021
print("Top 5 Countries on 2021-06-27 with the most vaccinations:")
print(str(top_5))
print("Bottom 5 Countries on 2021-06-27 with the least vaccinations:")
print(str(bottom_5))
pd.reset_option('display.float_format')

#task 4
#creating visualisation for top and bottom countries from task 1
#using line chart is the best visualisation for time series data
#the below lists are the top and bottom countries from task 1
top_countries = ['Brunei', 'Cuba', 'Chile', 'Portugal', 'Singapore', 'Hong Kong', 'Malta', 'South
Korea']
bottom_countries = ['Nigeria', 'Malawi', 'Zambia', 'Guinea', "Cote d'Ivoire", 'Kyrgyzstan',
'Russia', 'Croatia', 'Poland', 'Timor']

dataset_cleaned['date'] = pd.to_datetime(dataset_cleaned['date']) #converting date to datetime
format
dataset_cleaned['month'] = dataset_cleaned['date'].dt.to_period('M') #for representing month with
yearly level ie it adds month and year in output like Jan 2020
#checking the avg data of full vaccination per 100 by grouping months and country
monthly_vaccination = dataset_cleaned.groupby(['month',
'country'])['people_fully_vaccinated_per_hundred'].mean().reset_index()
```

```
#creating the range of months based upon our data
monthly_vaccination = monthly_vaccination[(monthly_vaccination['month'] >= '2020-10') &
(monthly_vaccination['month'] <= '2022-10')]
#creating function for line chart
def plot(countries, title):
    #checking data whether it is availabe with the montly_vaccination for those countries
    country_data = monthly_vaccination[monthly_vaccination['country'].isin(countries)]
    #its creates a table with month corresponding to country and vaccinations data per 100 people
    pivot_data = country_data.pivot(index='month', columns='country',
values='people_fully_vaccinated_per_hundred')
    #labeling
    ax = pivot_data.plot(figsize=(12, 8), title=title) #line chart title
    ax.set_xlabel("Month") #labeling x axis
    ax.set_ylabel("Fully Vaccinated People Per 100") #labelling y axis
    ax.set_xticks(pivot_data.index)  #sets tick marks on axis with pivot data
    ax.set_xticklabels(pivot_data.index.strftime('%b %Y'))  #sets month labels ie jan feb....
    plt.xticks(rotation=45) #sets rotation of month labels for avoiding overlapping
    plt.tight_layout()
    plt.show() #displays the line chart
#these below uses the above function and plots graph for top and bottom countries of task 1 over
2 years
plot(top_countries, 'Top Countries by Fully Vaccinated People (Per 100) Over Time (Oct 2020 -
Oct 2022)')
plot(bottom_countries, 'Bottom Countries by Fully Vaccinated People (Per 100) Over Time (Oct
2020 - Oct 2022)')
```